

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month, in Earth we learn about the beautiful 'mother of pearl' clouds which have a dark side, in Health we explore Coronavirus outbreaks, in Ignorance we tackle the problem of determinism, in Space Age we learn about the Lunar gateway, and finally, in R&D we delve in materials for future electronics



Biodegradable marine plastics

Chemists from Cornell University have developed a plastic for marine use that degrades faster when exposed to ultraviolet light, while preserving all the mechanical properties required for commercial fishing equipment. If the gear or nets are lost at sea, they degrade within months rather than decades.

<https://digit.in/ghostnets>

WHAT'S NEW

Contact tracing without compromising on privacy

Numerous universities are developing the technology necessary for contact tracing – a method that allows public health agencies to figure out if a person has been exposed to someone who has tested positive for the novel coronavirus, while ensuring that the privacy of the people are not compromised. University of Washington and Microsoft developed an app, CovidSafe, that stores all the data locally on the smartphone till a person tests positive. The individual can then relay the data back to the servers, after which it pings the anonymised data to everyone who has installed the app. Those who



may have been in contact with the affected individual get an alert. MIT is spearheading a multi-university project, Safe Paths, that uses distributed deep learning. Stanford students developed an app called Covid Watch that uses bluetooth chirps associated with pairing devices to trace contacts. The information does not leave the smartphone till an affected person intends to alert contacts of their positive status. Location data is not collected by the app at all, only proximity.

Why are bat viruses so deadly?

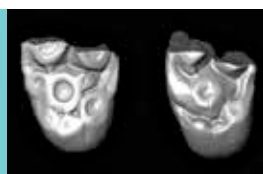
One of the questions raised by the novel coronavirus outbreak is exactly why bat viruses are so deadly. A number of recent outbreaks have their origin in bats, including SARS, Ebola, MERS, Marburg, Nipah, and, very likely, Covid-19. One of the reasons may be the strong immune systems of bats. While bats themselves might be good at fighting the viruses, the viruses evolve to become increasingly deadlier, and wreak



havoc when they jump to humans. The immune systems of the bats encourages the virus to reproduce more rapidly within the host organism, before there is time for the immune system to respond to an infec-

tion. One aggravating factor for spreading of viruses from bats to other animals is the disruption of their natural habitats. When bats are stressed, the viral loads they shed shoots up, which increases the chances of cross species infections. The Bat One Health project funded by DARPA is conducting research to track how disruptions in bat habitats causes disease to spread among other animals.

<https://digit.in/batvirus>



Monkeys crossed the Atlantic

Fossilised teeth of monkeys found in South America, shows that some ancient primates managed to survive the trip across the Atlantic all the way from Africa, probably on floating vegetation that was dislodged from coastal areas following tropical storms.

<https://digit.in/mnkytIntc>



Alerts for stressed plants

MIT engineers have developed nanosensors made up of carbon nanotubes that can be coated on leaves of plants. When the leaves experience any kind of injury, infection or damage, they can send an alert to smartphones. The sensors can be used at scale in agriculture.

<https://digit.in/plantsensors>



Hubble celebrates birthday

The Hubble Space Telescope celebrated it's 30th birthday with another spellbinding image of the cosmos. The image captures the giant nebula NGC 2014 and the neighbouring NGC 2020 which are both nurseries for new stars in the Large Magellanic Cloud.

<https://digit.in/starbirth>